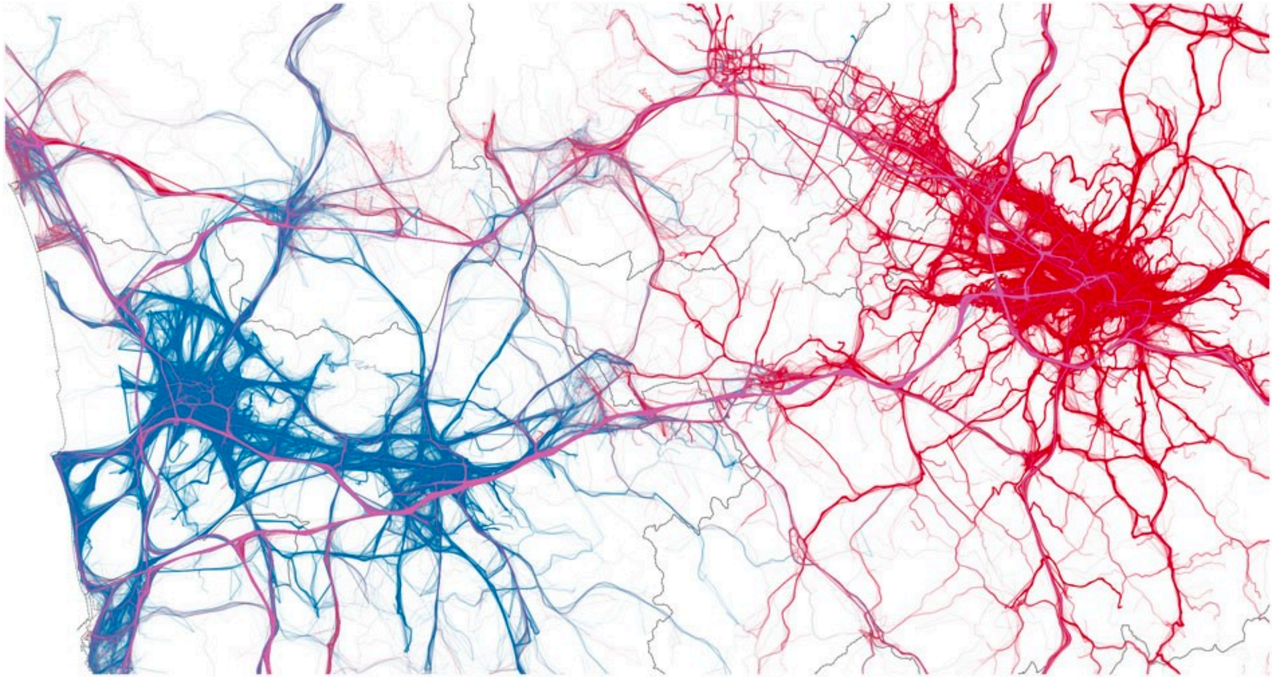
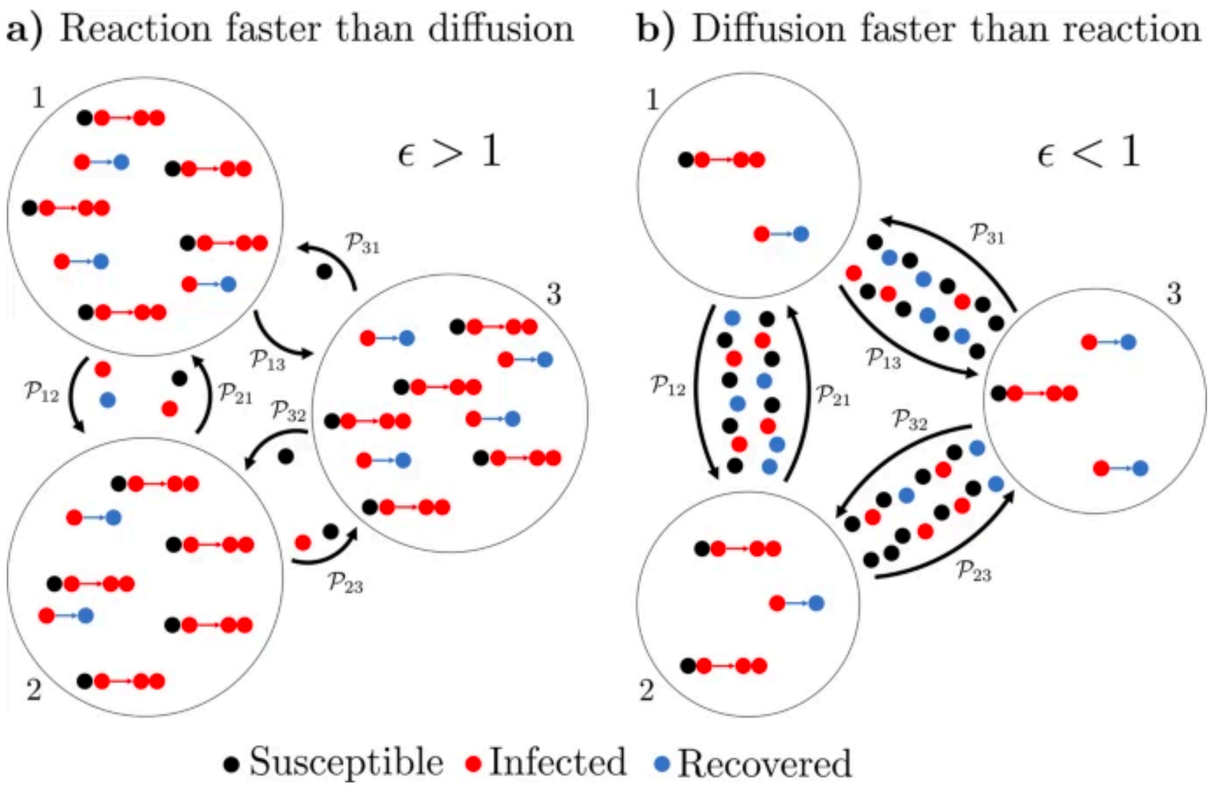
Human flows and contagion

Course: Physics of Complex Networks: Structure and Dynamics





Objective: compare epidemics from data vs human flow models

$$\dot{S}_i = -\beta_i S_i(t)\frac{I_i(t)}{N_i(t)} + \frac{1}{\epsilon}\left[\sum_j \Phi_{ji}S_j(t) - \sum_j \Phi_{ij}S_i(t)\right]$$

$$\dot{I}_i = \beta_i S_i(t)\frac{I_i(t)}{N_i(t)} - \gamma I_i(t) + \frac{1}{\epsilon}\left[\sum_j \Phi_{ji}I_j(t) - \sum_j \Phi_{ij}I_i(t)\right]$$

$$\dot{R}_i = \gamma I_i(t) + \frac{1}{\epsilon}\left[\sum_j \Phi_{ji}R_j(t) - \sum_j \Phi_{ij}R_i(t)\right]$$

$$N = \sum_{i=1}^{M} N_i(t) = \text{const} \qquad N = \sum_{i=1}^{M} N_i(0)$$

$$N_i(t) = S_i(t) + I_i(t) + R_i(t)$$

Settings & sanity checks

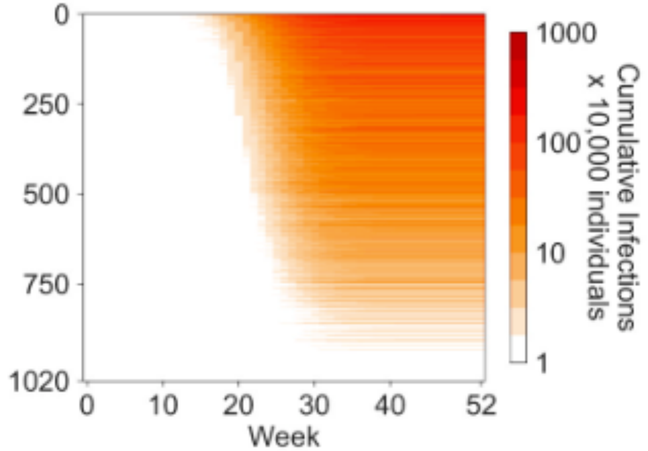
Pro hint: $\vec{x} \equiv (S_1, S_2, \ldots, S_M, I_1, I_2, \ldots, I_M, R_1, R_2, \ldots, R_M)$

Castioni, Gallotti, De Domenico, Nature's Communications Physics (2021)

(1) Write the code to solve the ODE (re-adapt the structure of the code used for Kuramoto)

(2) Initial conditions (t=0):

- Use population data from Lab Session 5 to set the number of susceptible (S_i) in each patch
- Assign 20 infected individuals (I_i) to Padua
- Use $\beta_i = \beta = 0.5$ $\quad \gamma = 0.2$ $\quad \epsilon \in \{1, 10^2, 10^4\}$

(3) Integrate for 365 time steps and visualize a comparison of data vs gravity vs radiation model (4 panels):

- For each ϵ : Heatmap of $\frac{I(t)}{N}$ (x axis: time; y axis: node ID)
- What is the impact of mobility matrices on the final outcome?